

Visual Integrity in the Age of AI: An Evaluation of DLSS and DLAA in Geospatial Visualization

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ABSTRACT

Deep Learning Super Sampling (DLSS) is a relatively new machine learning technology for accelerating 3D graphics, by rendering at lower resolutions, and inferring what the higher resolution output should look like. However, DLSS was designed for, and trained on, video games. Can this technology, intended for the entertainment market, be repurposed to accelerate high-resolution scientific and geospatial visualization? This paper evaluates the visual accuracy costs of inferring higher resolution data visualizations with DLSS upscaling. A related technology, Deep Learning Anti-aliasing (DLAA) is similarly evaluated. This evaluation focused on 3D geospatial applications, with (primarily non-photorealistic) terrain rendering and point cloud rendering, though the results are applicable to a wider range of scientific visualization data types. The results demonstrate that DLSS/DLAA can significantly impact visual accuracy, and should be avoided for visualizations requiring accurate portrayal of fine details.

1 INTRODUCTION

Rendering resolutions for scientific and geospatial visualizations are continually increasing. Driving factors include the increasing detail of remote sensing and modelled data, and the adoption of ultra-high-resolution displays. Virtual reality (VR), in particular, requires both high resolutions and high frame rates (to avoid simulator sickness). Achieving these demanding performance targets often requires compromises in visual detail and/or prohibitively expensive GPUs.

In recent years, upscaling technologies such as DLSS[5], FSR, and XeSS have become widely adopted in the video gaming industry to improve performance by rendering fewer pixels and then using AI-driven upscalers to reconstruct missing details. These upscaling technologies were developed as successors to traditional anti-aliasing (AA) techniques, offering both performance benefits and image quality improvements. Deep Learning Anti-Aliasing (DLAA) employs the same neural network architecture as DLSS, but without resolution scaling, operating at native resolution to provide AA. This paper focuses on NVIDIA's DLSS/DLAA technology, as it is the most mature and widely implemented solution.

DLSS has proven effective in improving performance [10] in video games, where maximizing frame rates justifies reductions in visual fidelity. However, in scientific and geospatial visualization, visual integrity and accuracy are critical, with frame rates generally being a secondary concern (with the notable exception of VR).

When visualizing geospatial data such as terrain and bathymetry (underwater depth measurement) surfaces, alterations introduced by

AI-upscaling algorithms (like DLSS) can be problematic. Potential issues include:

- **Geometric Distortion** of fine shape/shading details, or elements such as contour lines, can alter perceived surface shape or feature boundaries.
- **Color Alteration:** Color often encodes scalar/categorical values, thus shifts in color can cause perception of incorrect data values.
- **Feature Hallucination/Deletion:** The inferential nature of DLSS can introduce details not present in the data (hallucinations) or erroneously remove small, valid features.

These issues are potentially exacerbated by DLSS being primarily trained on video game recordings, not scientific or geospatial visualizations. As such, it is possible that they may not perform as well when used on content that is uncommon in games, e.g. surfaces with rainbow color maps, or point cloud data.

To our knowledge, this paper presents the first quantitative evaluation of DLSS and DLAA applied specifically to geospatial scientific visualizations. While some previous studies have assessed DLSS quality in gaming environments [9], these studies typically focused on photorealistic scenes and considered optimization for entertainment rather than scientific accuracy. Our research addresses this gap by providing a quantitative evaluation in order to help researchers and visualization professionals make informed decisions about incorporating DLSS/DLAA technology into scientific/geospatial visualization pipelines.

2 METHODOLOGY

Test scenes were created to represent common 3D geospatial content: Multiple 2.5D terrain and bathymetry surfaces were created using raster height map Digital Elevation Models and Bathymetric Attributed Grids, rendered as triangle strips. Depending on the test, these were color-mapped by height/depth, with and without contour lines, or had photorealistic textures.

More-complex 3D surface models were represented by a 15 million triangle model of a coral reef, generated using SfM (Structure-from-Motion), colored with rugosity values [3]. Point clouds were represented using a SfM reconstruction of a coral reef, and a LIDAR scan of Tokyo [12]. Points were rendered with and without eye-dome lighting [4][11].

For each scene, multiple screenshot images were taken at set frames along predetermined camera motion paths, with many frames in between each. Rendering conditions were:

- Reference images, rendered at 4K (3840x2160) with *No-AA*, and at 8K (7680x4320) using Super Sampling Anti-Aliasing (*SSAA*).
- AA test images, rendered at 4K (3840x2160) using *DLAA*, Fast Approximate AA [7] (*FXAA*), and Temporal AA [14] (*TAA*).
- *DLSS* test images, rendered at reduced resolutions determined by NVIDIA's preset modes: *Quality* (67%, 2560x1440), *Balanced* (58%, 2227x1280), *Performance* (50%, 1920x1080), and *Ultra-Performance* (33%, 1280x720), then upsampled to 4K (3840x2160) using NVIDIA's official DLSS 4 (2025.02.25_UE_5.5 Plugin_4.0.0) plugin for Unreal Engine 5.5.4.

The differences between reference images and test condition images were analyzed using multiple image quality metrics. Based on a survey of image quality methods for compressed images [6], Butteraugli [1] and Multi-Scale Structural Similarity Index (MS-SSIM) [13] were chosen as complimentary standardized comparison

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metrics. Butteraugli assesses psycho-visual similarity, focuses on just-barely noticeable differences, and provides heat maps of differences that highlight the distribution of upscaling artifacts (see supplemental materials for example difference maps). MS-SSIM considers structural information, provides information on the size distribution of upscaling artifacts, and was noted to be effective with AI compression artifacts [6]. NVIDIA's FLIP [2] was also used, as it was designed specifically for comparing rendered images to ground truth rather than analyzing compression artifacts, and it addresses shortcomings of previous algorithms such as Butteraugli and MS-SSIM. We primarily used FLIP's single weighted median value.

Surfaces are often color-coded to indicate particular values, thus if upscaling artifacts change colors, incorrect values will be perceived. To assess this, images were converted into HSV color space, and their hue values were compared. Hue differences were thresholded to create a binary image of hue differences greater than 3%. Using a 3x3 structuring element, morphological closing was used to connect nearby single pixel differences into artifact clusters, then morphological opening was used to eliminate clusters thinner than three pixels wide. The number of remaining artifact clusters and the percentage of screen pixels they occupied was recorded.

Shading provides strong visual cues of surface shape. To understand how upscaling artifacts could distort perception of surface shape, images were converted to CIELAB color space, and differences in their luminance values greater than 5% were analyzed using the same method previously described for hue.

3 RESULTS

3.1 Impact on Quality Metrics Using AA as Baseline

To help evaluate the impact of DLSS on visualization accuracy, we established a baseline for our comparison metrics by also evaluating the image changes introduced by common AA techniques:

- **Fast Approximate Anti-Aliasing (FXAA)** identifies edges in the final rendered image and smooths them, offering extremely high performance, but often blurs textures and text.
- **Temporal Anti-Aliasing (TAA)** accumulates information across multiple frames, using motion vectors to align previous frames with the current one, providing good quality with minimal performance impact, but potentially introducing ghosting artifacts during rapid movement and blurring fine details like thin lines.
- **Super Sampling Anti-Aliasing (SSAA)** renders at higher resolution before downsampling; produces superior image quality at significant computational expense.
- **Deep Learning Anti-Aliasing (DLAA)** DLSS without upscaling, to provide enhanced AA with better preservation of fine details compared to TAA, though at a higher performance cost than TAA.

Figure 1 shows the impact of these AA algorithms on image quality.

Due to significant differences in observed behavior, we analyzed the six 2.5D & 3D surface mesh scenes separately from the four point cloud scenes.

Table 1 presents the mean image comparison metric (Butteraugli, FLIP, and MS-SSIM) scores for scenes with surface meshes, for each AA and DLSS condition, with No-AA and SSAA as reference images.

Table 1. Mean image comparison scores for surface mesh scenes. Lower Butteraugli and FLIP values indicate more similarity, higher MS-SSIM values indicate more similarity. Note DLSS scores progressively worsen as rendering resolution decreases from 66% to 33%.

AA method	Butteraugli		FLIP W. Median		MS-SSIM	
	NoAA	SSAA	NoAA	SSAA	NoAA	SSAA
FXAA	13.21	22.72	0.057	0.097	0.92	0.78
TAA	27.60	24.04	0.074	0.099	0.82	0.76
SSAA	24.53	-	0.095	-	0.79	-
DLAA	25.08	19.93	0.057	0.091	0.80	0.77
DLSS 66%	30.03	25.85	0.077	0.116	0.79	0.75
DLSS 58%	31.31	27.39	0.089	0.126	0.77	0.74
DLSS 50%	33.74	30.85	0.102	0.137	0.76	0.72
DLSS 33%	41.69	40.66	0.141	0.167	0.72	0.68

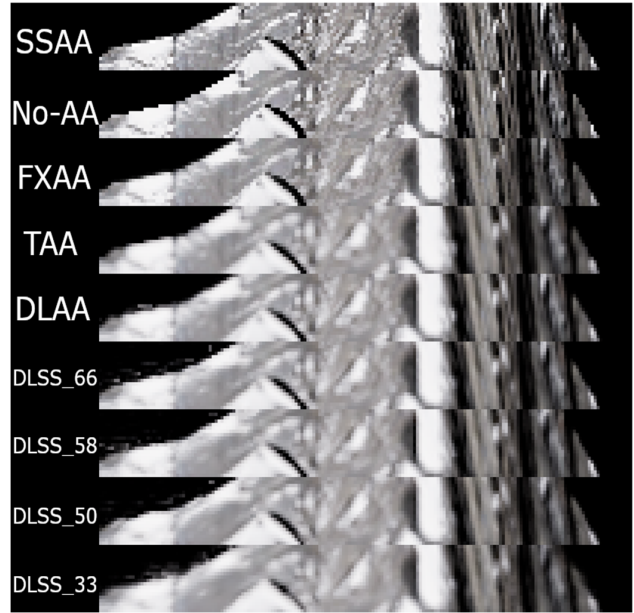


Fig. 1 Comparison of AA and DLSS methods applied to mountainous terrain with realistic textures. Note the differences in edge treatment along ridgelines and texture preservation in the snow and rock textures.

As shown in Table 1, the various AA methods introduced different degrees of visual change compared to the No-AA reference. FXAA introduces the least deviation, followed by DLAA, then TAA, with SSAA trailing behind. It is important to note that these metrics quantify the degree to which visual changes are noticeable, rather than the subjective quality of the AA. Thus, while SSAA introduced more noticeable changes, these changes were based on a higher-resolution rendering of real data.

Considering all of the metrics together, DLSS introduced more-noticeable image changes than most common AA methods. In all cases, DLSS showed progressively worsening visual differences as rendering resolution decreased, with DLSS 33% (Ultra Performance preset) performing the worst across all metrics.

When SSAA was used as a reference image, DLAA demonstrated the closest visual similarity among all tested AA algorithms, though it should be noted that FXAA showed a slightly better MS-SSIM score. However, considering all metrics together, DLAA provides the most consistent approximation of SSAA's visual characteristics while requiring significantly less computational resources.

3.2 Visual Artifacts

These metrics demonstrate that DLSS introduces noticeable visual changes; but to understand if these changes can affect perception of surface shape or encoded values, we consider their impact on hue and luminance values. Table 2 presents the percentage of screen pixels in significant hue or luminance artifact clusters (as defined in Section 2) across all scenes with surface meshes, using SSAA as the reference.

Table 2. Percentage of screen pixels in significant hue and luminance artifacts clusters, for surface mesh scenes (compared with SSAA). Lower percentages indicate fewer visual artifacts.

AA method	% of pixels in clusters of Hue Differences >3%	% of pixels in clusters of Luminance Differences >5%
FXAA	4.56	3.604
TAA	4.78	5.796
DLAA	4.67	4.853
DLSS 66%	4.95	6.511
DLSS 58%	5.05	7.301
DLSS 50%	5.18	8.178
DLSS 33%	5.55	10.069

DLAA performed in between FXAA and TAA, indicating it can likely be used in visualizations without major concern. DLSS, however, resulted in more-significant hue and luminance artifacts than AA, with increasing artifact presence as rendering resolution decreased. Compared to AA methods, DLSS showed a modest increase in hue artifacts (+6% to +19%), and a steep increase in luminance artifacts (+39% to +214%). This indicates that DLSS has considerable potential to impact accurate perception of surface shape through distorted shading cues. Example artifacts are shown in Figure 2:

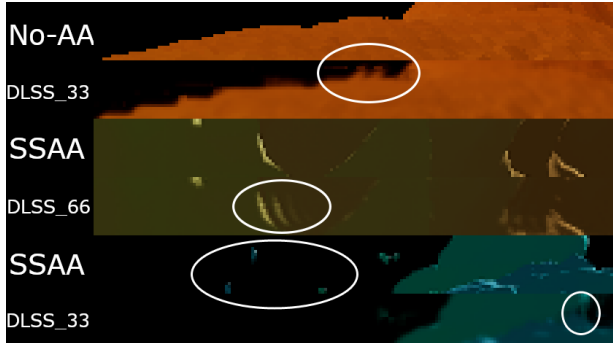


Fig. 2 DLSS artifacts: (top) hallucinated geometry, (middle) ghosting, (bottom) lost geometry and hallucinated holes.

As the test surfaces increased in complexity (from relatively-flat seafloor, to more-rugged terrain, and finally an intricately detailed coral reef), DLSS and DLAA generally produced progressively more noticeable visual differences. Beyond geometric complexity, differences in DLSS output are also driven by textural complexity.

3.3 Texture

To understand the impact on finely detailed structured textures (e.g. grid or contour lines) often used to reinforce perception of surface shape, we tested terrain surfaces with and without contour lines. The addition of contour lines worsened mean DLSS Butteraugli scores by 53%, FLIP 338%, and MS-SSIM 9%. Much of this can be attributed to the thickening of the lines; relative to the surface underneath, the visual weight of one-pixel wide lines increases as rendering resolution decreases (see Figure 3). DLAA also showed similar degradation of scores (39%, 82%, and 4% worse). To exclude line width effects, we compared DLAA against No-AA, which renders at the same resolution and thus produces the same relative line width. Scores still worsened, suggesting that the degradation is not solely attributable to line width/resolution issues.



Fig. 3 One-pixel wide contour lines get thicker with DLSS upscaling.

We hypothesized that, as DLSS models were trained on more-photorealistic video game imagery, they could perform poorly on surfaces with color mapped values. We tested this by rendering a mountain scene (Figure 4), once with photorealistic grass, rock, and snow textures, and once with a height-based color gradient (Turbo[8]).

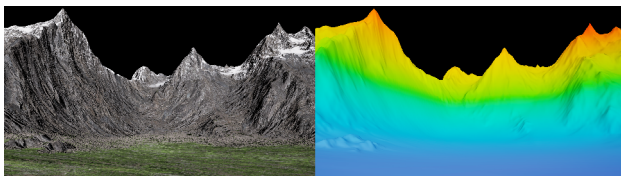


Fig. 4 Terrain, with photorealistic textures, and colored by height.

The test showed that DLSS actually performed far better on the height-colored terrain compared to the photorealistic, with 16% to 48% improvements in Butteraugli, FLIP, and MS-SSIM scores, and a 99% reduction in significant hue and luminance artifact clusters. This can be attributed to the photorealistically textured scene having vastly more fine details that could be corrupted by DLSS.

3.4 Impact on Point Cloud Rendering

For point cloud scenes, measuring image quality for AA methods and DLSS presented unique challenges, as point clouds exhibit fundamentally different behavior compared to surface meshes when rendered at different resolutions. We used single-point (single pixel) rendering rather than squares or circles, as this approach is significantly faster and allows for the display of more individual points onscreen without occlusion. (Applications often require rendering hundreds of millions of points simultaneously.) Figure 5 illustrates how techniques rendering at higher and lower resolutions can dramatically affect point cloud visualization, influencing apparent individual point size, overall visual density, and occlusion issues:

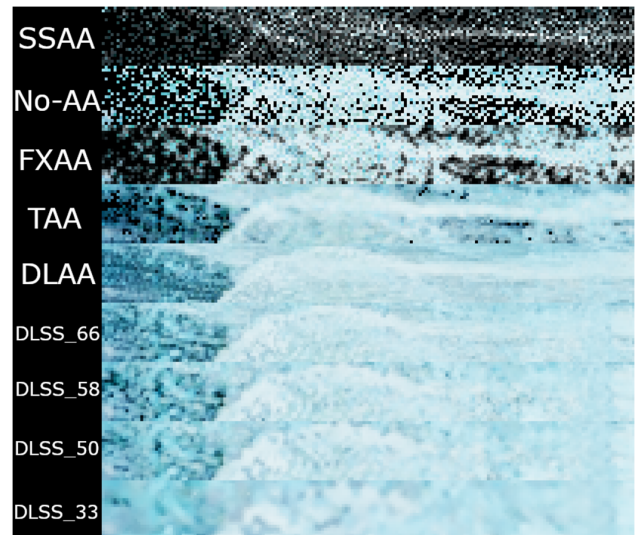


Fig. 5 Visual comparison of various AA/DLAA/ DLSS techniques on point cloud rendering, demonstrating the effects of rendering resolution on point visualization.

Due to these unique behaviors, the metrics in Table 3 show extreme values, reflecting the substantial visual differences that result from applying AA and DLSS methods to point clouds. The SSAA results in particular indicate that point cloud representations at different resolutions become visually so distinct that direct comparison is challenging.

Table 3. Mean image comparison scores for point cloud scenes. Note the extreme values across all metrics, indicating that point cloud representations are fundamentally altered by AA/DLSS methods.

AA method	Butteraugli		Flip W. Median		MS-SSIM	
	No-AA	SSAA	No-AA	SSAA	No-AA	SSAA
FXAA	41.75	38.50	0.500	0.409	0.72	0.80
TAA	50.56	62.09	0.519	0.652	0.40	0.41
DLAA	52.05	79.77	0.410	0.805	0.42	0.36
DLSS 66%	64.46	106.51	0.417	0.916	0.45	0.27
DLSS 58%	68.06	117.01	0.412	0.934	0.41	0.24
DLSS 50%	66.29	129.51	0.411	0.944	0.37	0.20
DLSS 33%	115.06	172.87	0.793	0.961	0.11	0.10

Based on these results, it is not recommended to use most AA algorithms with single-point rendering, only when points are rendered as larger shapes. However, if larger points are desired, DLSS can be

used with single-pixel rendering to achieve a similar visual result with faster performance. For example, Figure 6 shows how upscaling single-pixel points with DLSS 33% approximates full-resolution rendering with points sized 3 pixels wide.

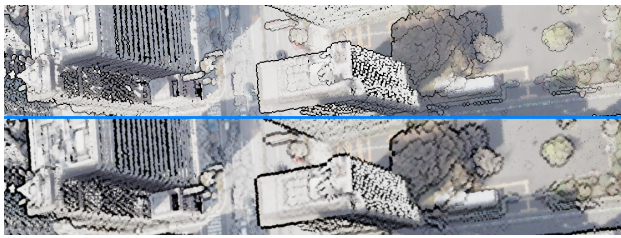


Fig. 6 Point clouds with eye-dome lighting, (top) full-res rendering with point size=3 pixels, (bottom) DLSS 33% with point size=1 pixel.

Eye-dome lighting[4][11] is often used on point clouds to improve depth perception by adding subtle shading and shadowing effects to distinguish features that would otherwise blend together. As shown in Figure 7, DLSS/DLAA produced significantly worse Butteraugli, FLIP, and MS-SSIM scores when eye-dome lighting was enabled.

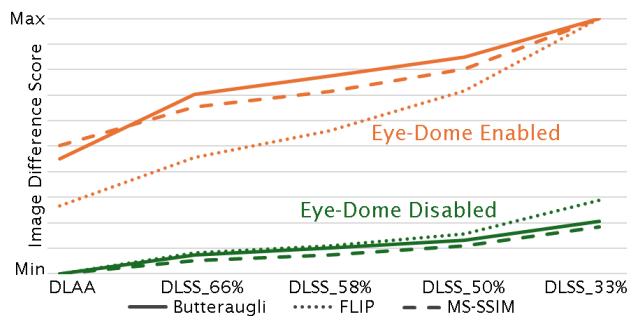


Fig. 7 Image comparison scores for DLSS/DLAA on point clouds rendered with and without eye-dome lighting enabled. For visualization purposes, Y-axis values were normalized between their min and max observed values, with Butteraugli scores scaled from the range [29.04, 117.94], FLIP scores from [0.40, 0.82], and MS-SSIM scores from [0.10, 0.74].

4 FUTURE WORK

This evaluation utilized the CNN-based DLSS model. Recently, NVIDIA released their transformer-based DLSS architecture, which represents an evolution in upscaling technology. Preliminary testing suggests that the transformer model demonstrates different shape preservation capabilities compared to the CNN approach, as illustrated in Figure 8. However, a more in-depth analysis should be conducted to determine which model is better for visualization applications.



Fig. 8 Shape preservation comparison, showing SSAA reference (top), CNN-based DLSS (middle), and transformer-based DLSS (bottom), demonstrating transformer's superior resemblance to SSAA.

Limited image quality analysis on a terrain mesh scene showed mixed results, with some metrics improving (Butteraugli) and some worsening (FLIP). Although the transformer architecture shows promise, the quantitative differences do not appear dramatic enough to conclude that this new version of DLSS will substantially mitigate the visual artifacts and accuracy concerns documented here.

Ideally, a new DLSS model could be fine-tuned by NVIDIA specifically for scientific visualization to complement the existing presets available in DLSS.

The evaluations presented in this paper would also benefit from expanded testing with additional scenes that are representative of scientific visualization domains beyond geospatial visualization. Furthermore, future work should extend this evaluation to AMD's FSR and Intel's XeSS. FSR is particularly interesting as pre-v4 versions used non-AI approaches with broad GPU compatibility.

5 CONCLUSIONS

Our evaluation was based on three complimentary image comparison metrics, and quantitative analysis of perceptually significant hue and luminance artifact clusters. The metrics showed that in almost all cases, DLSS produced significantly more-noticeable changes than common anti-aliasing techniques (a baseline for acceptable changes).

DLSS upscaling only modestly increased hue artifacts, indicating it is unlikely to seriously affect perception of values encoded by color, though it can hallucinate small, wrong-color artifacts near edges.

DLSS substantially increased the size and number of clusters of significant luminance artifacts. These luminance artifacts can distort surface shading details that provide important depth cues, leading to inaccurate perception of surface shape. This, combined with hallucinated geometry and ghosting artifacts, strongly suggests DLSS should not be used when accurate portrayal of fine details is critical.

The visual artifacts observed in this study can have a profound negative impact on inspection and detection tasks, e.g. in the processing of raw remote sensing data into finished products. When cleaning raw lidar/sonar point clouds, artifacts in point rendering can distract the user (false positives) and obscure the bad data points they are searching for (false negatives). The inspection of derived surface products often requires extreme accuracy, as any overlooked defects can have safety implications in products used for navigation (e.g. bathymetric surveys).

For point cloud rendering, DLSS should generally be avoided, as it lowers the number of distinct pixels that can be visualized without occlusion. An exception is when points need to be displayed larger than one pixel (e.g. to look more "solid"); in this case, single-point rendering at a reduced resolution and then upscaling with DLSS can produce a similar visual effect at considerably lower rendering cost.

For surfaces, DLAA performed similarly to TAA and FXAA in approximating high-quality SSAA, and introduced fewer hue and luminance artifacts than TAA. Thus, DLAA may be an acceptable choice for applications that can tolerate minor artifacts in exchange for superior AA quality. It should be noted that DLAA, like TAA, has a temporal component that helps with temporal stability of AA results but also introduces temporal artifacts such as ghosting. But it should be avoided for point clouds, and used cautiously with fine contour or grid lines. The safest AA option is higher resolution rendering (SSAA) without post-processing, except when the rendering outcome directly depends on pixel size, e.g. single-pixel point rendering.

DLSS is obviously effective for accelerating 3D rendering. However, this evaluation demonstrated that it can have significant effects on visual accuracy. This tradeoff is acceptable for video games, but not for visualizations that require accurate portrayal of fine details. Fine-tuning a visualization-specific upscaling model could potentially address these shortcomings, but currently the visual artifacts introduced by AI-upscaling outweigh any productivity benefits of higher-resolution rendering.

SUPPLEMENTAL MATERIALS

The UE project with test scenes, scripts, and results data are available here: https://vislab-ccom.unh.edu/downloads/DLSS_Paper_Data.zip

ACKNOWLEDGMENTS

This research was supported by NOAA Grant NA20NOS4000196.

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